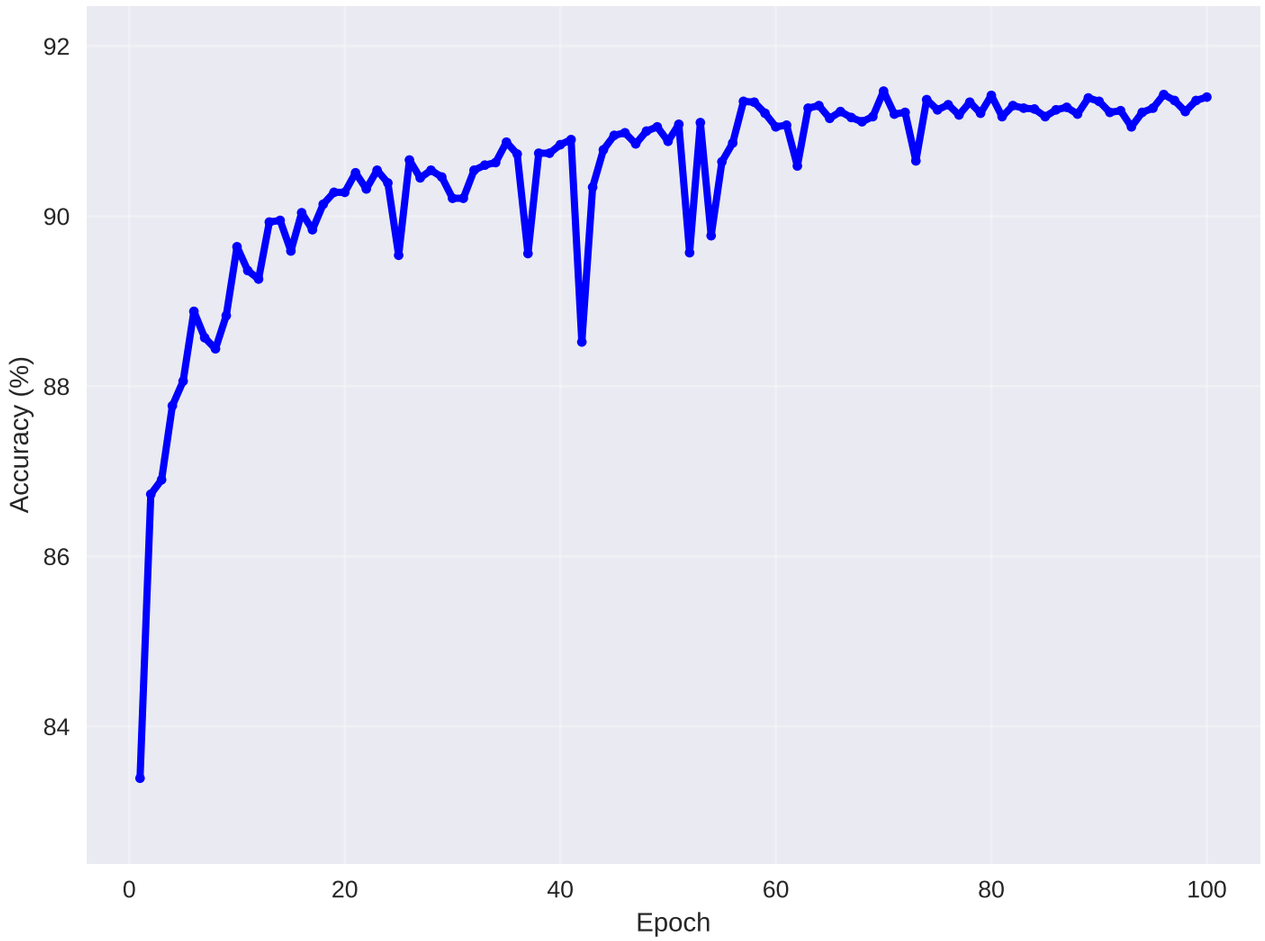
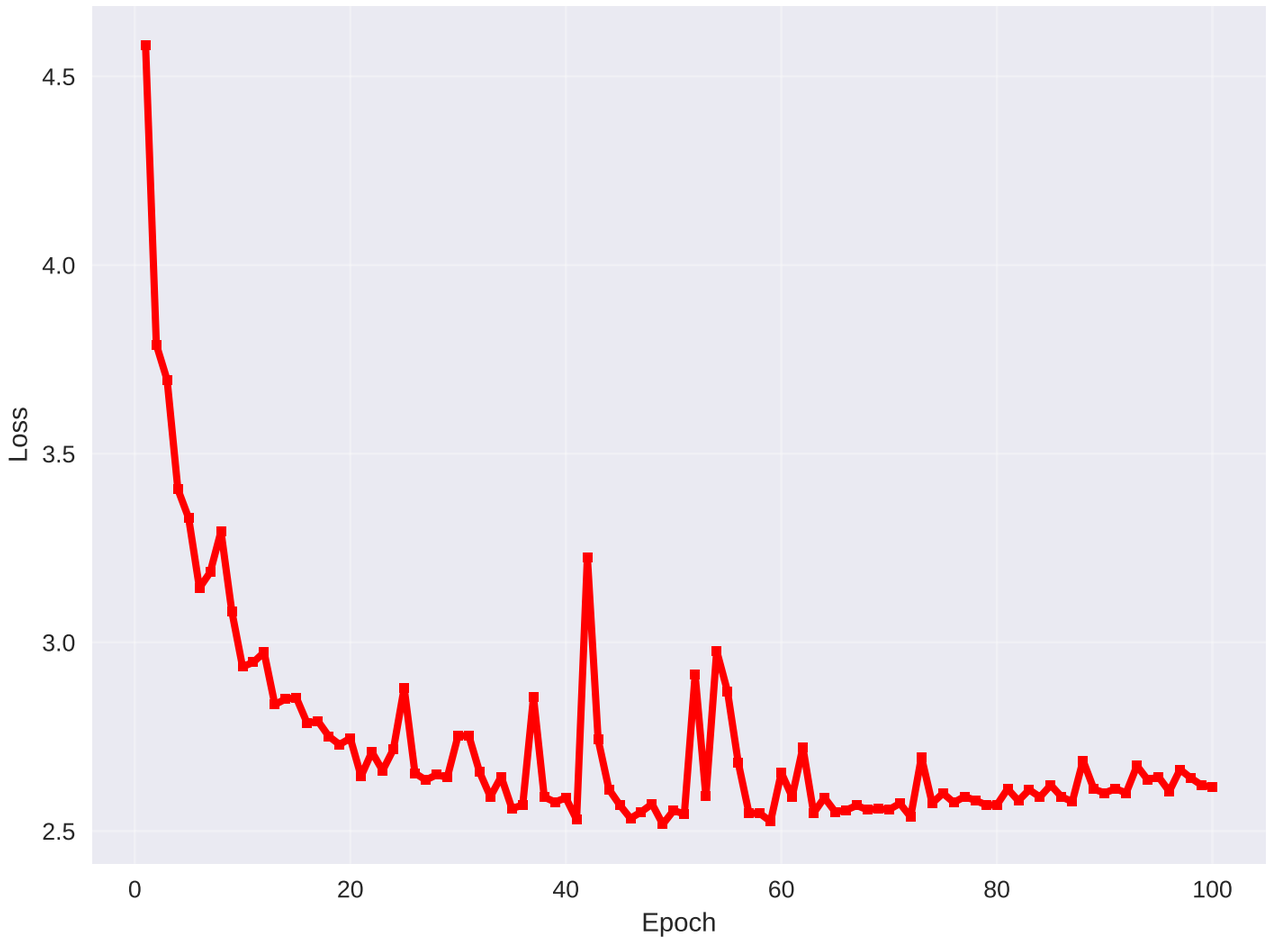
Federated Learning - Main Metrics Overview

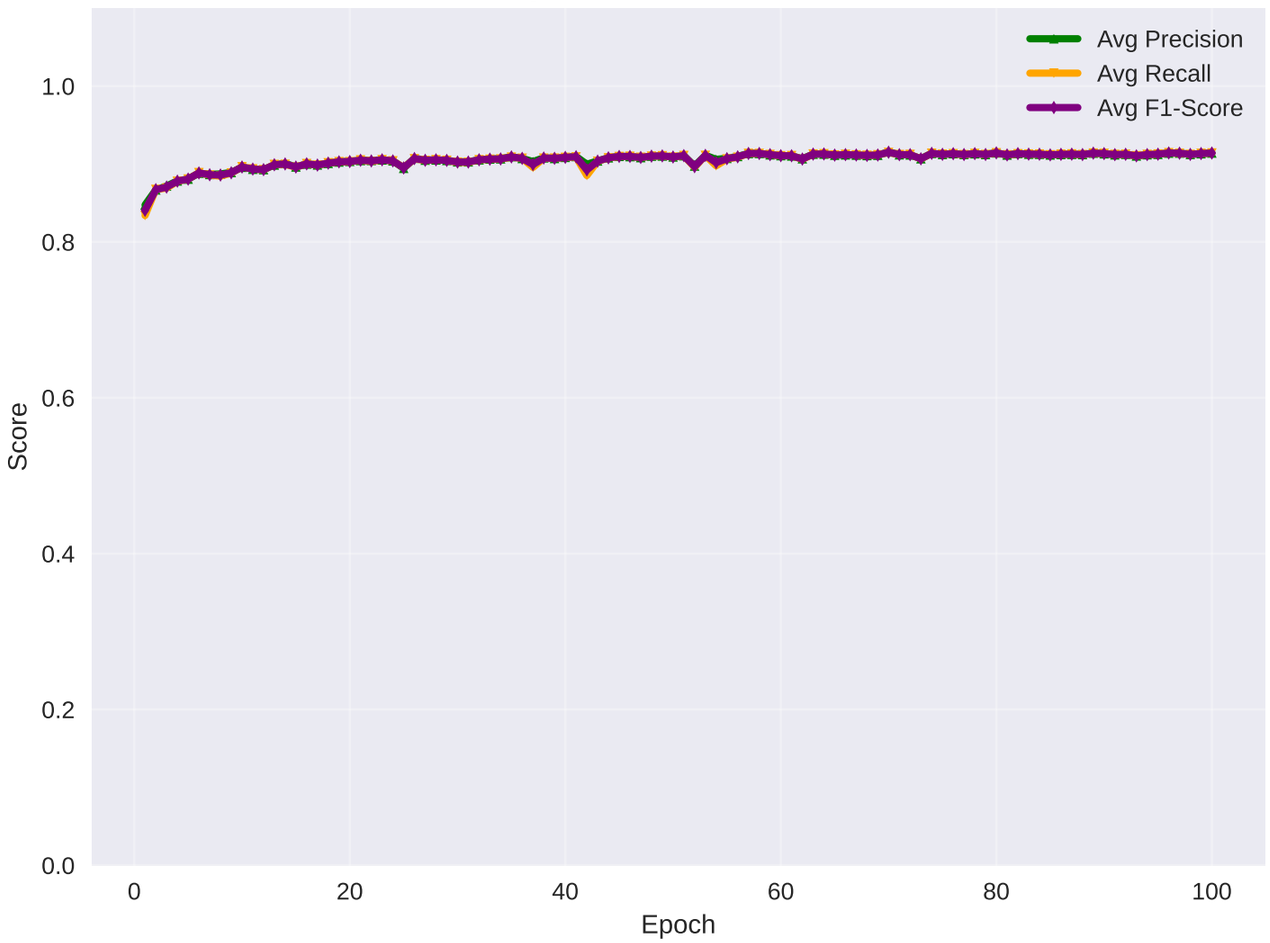
 Global Model Accuracy Evolution



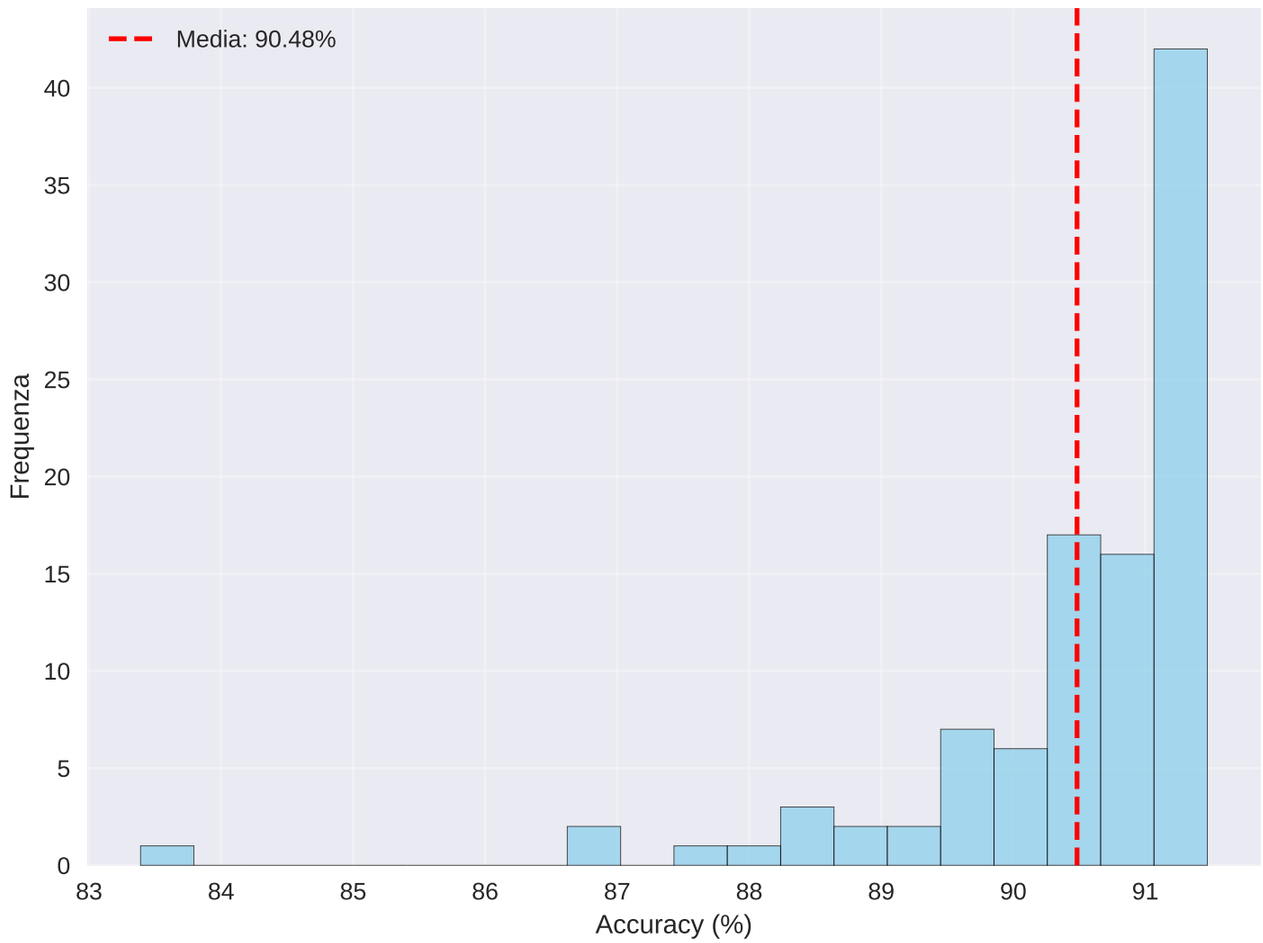
 Global Model Loss Evolution



 Average Precision, Recall & F1-Score

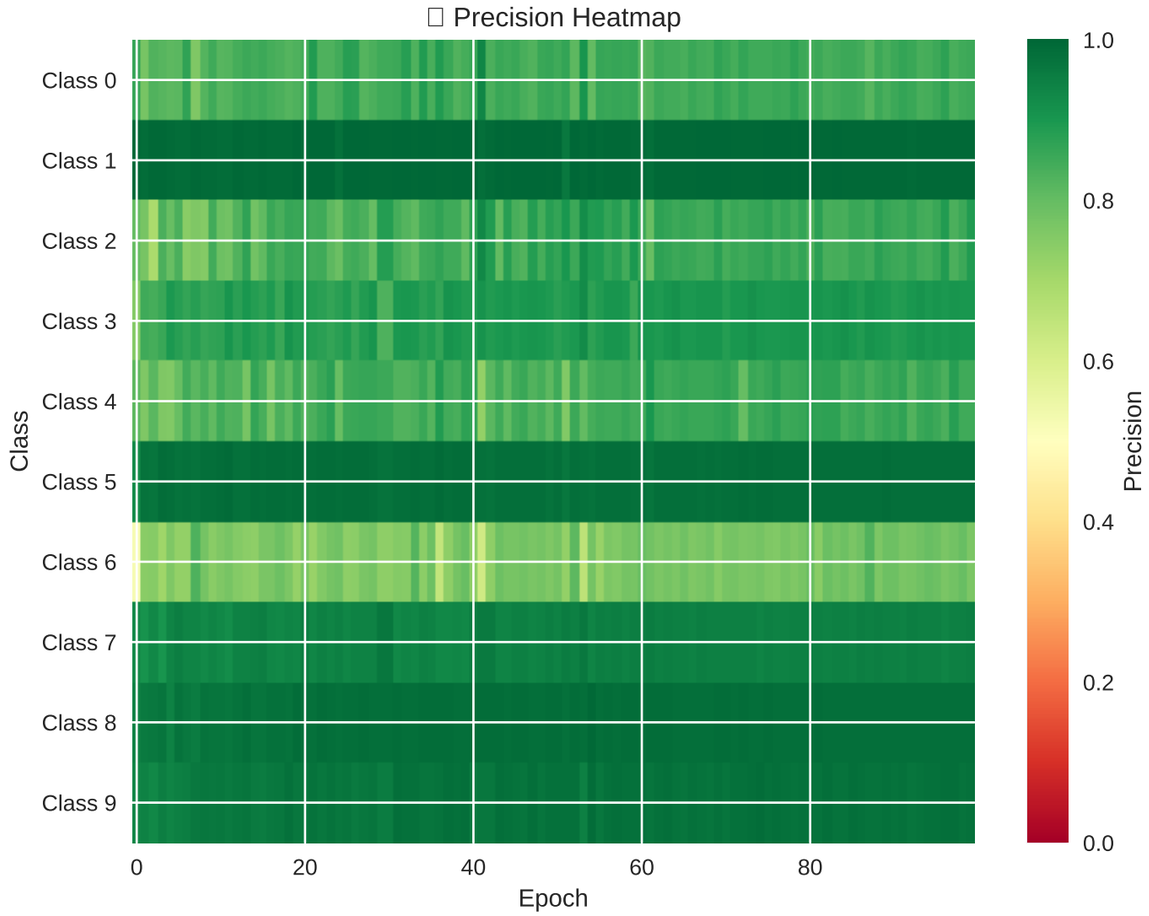


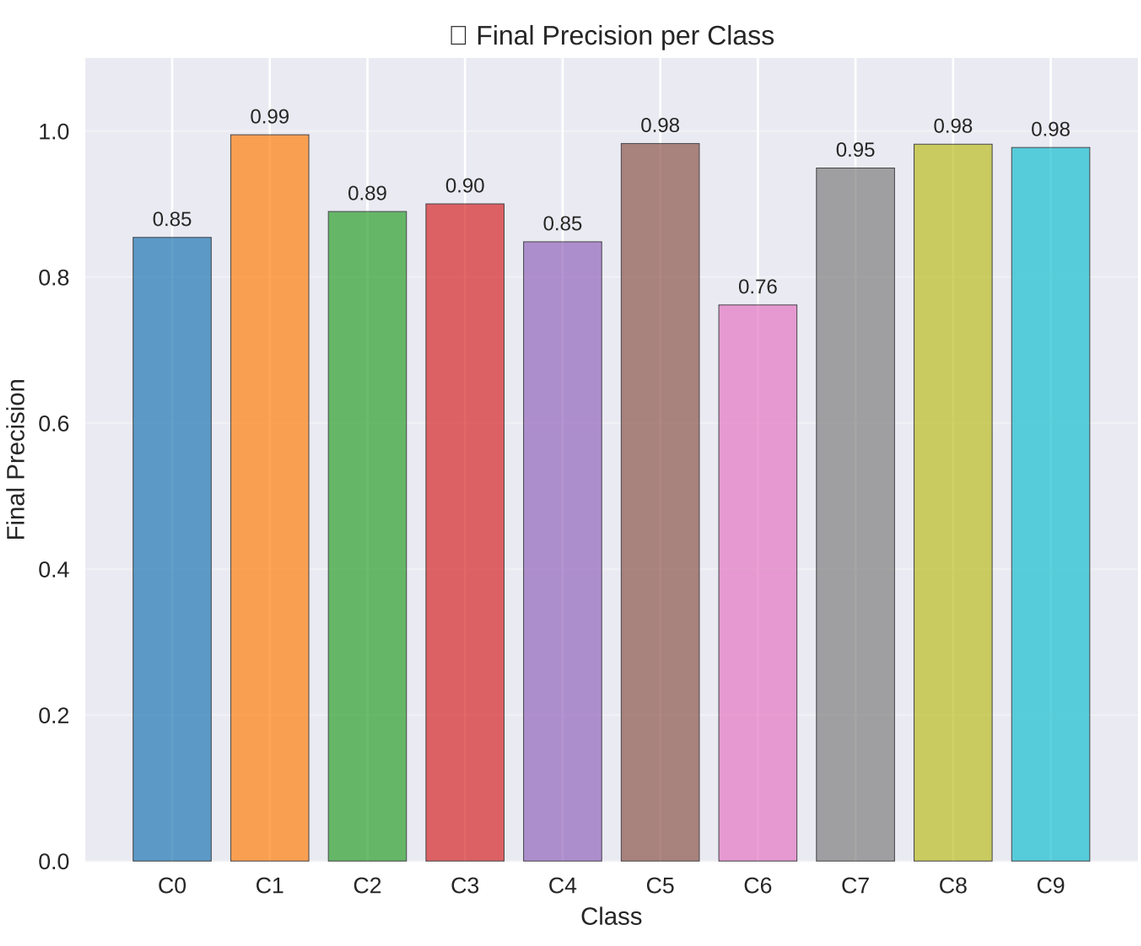
 Distribuzione Accuracy

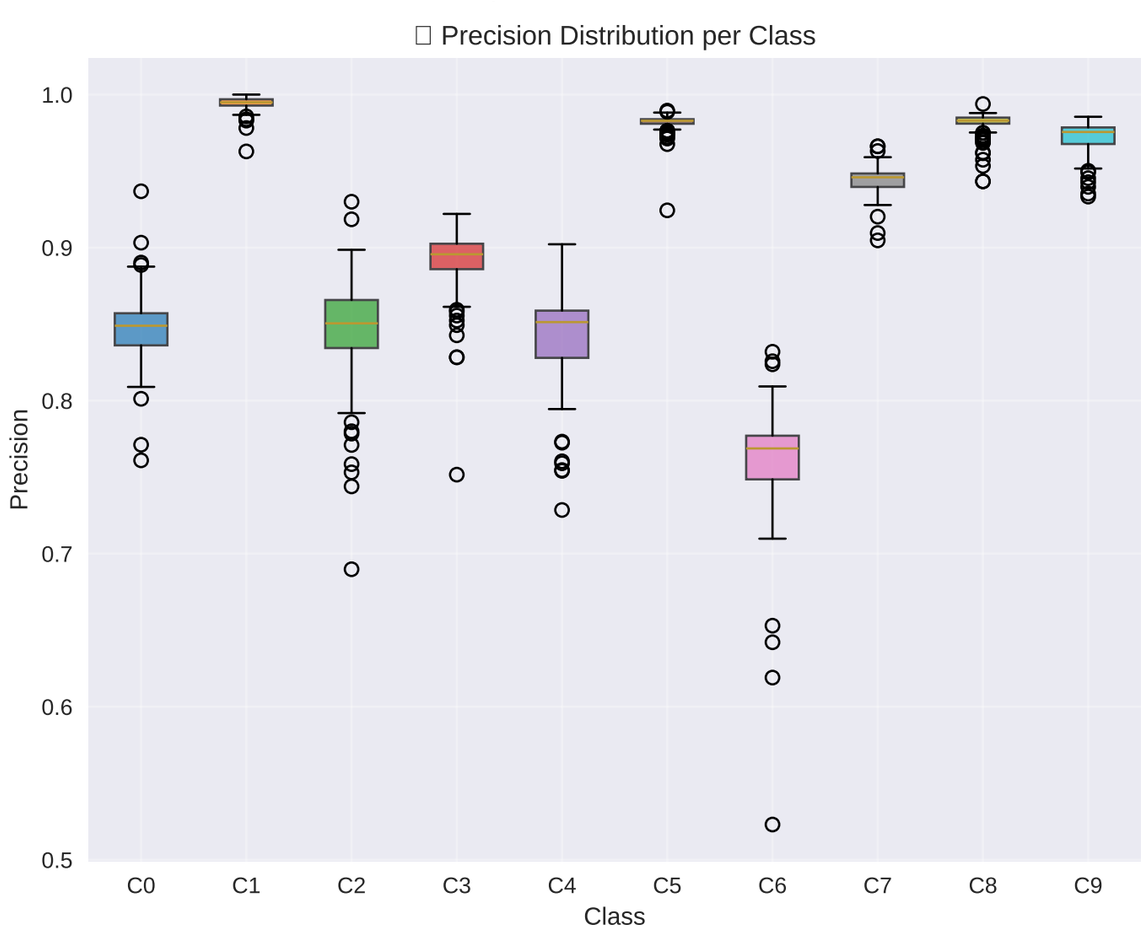


Per-Class Precision Analysis



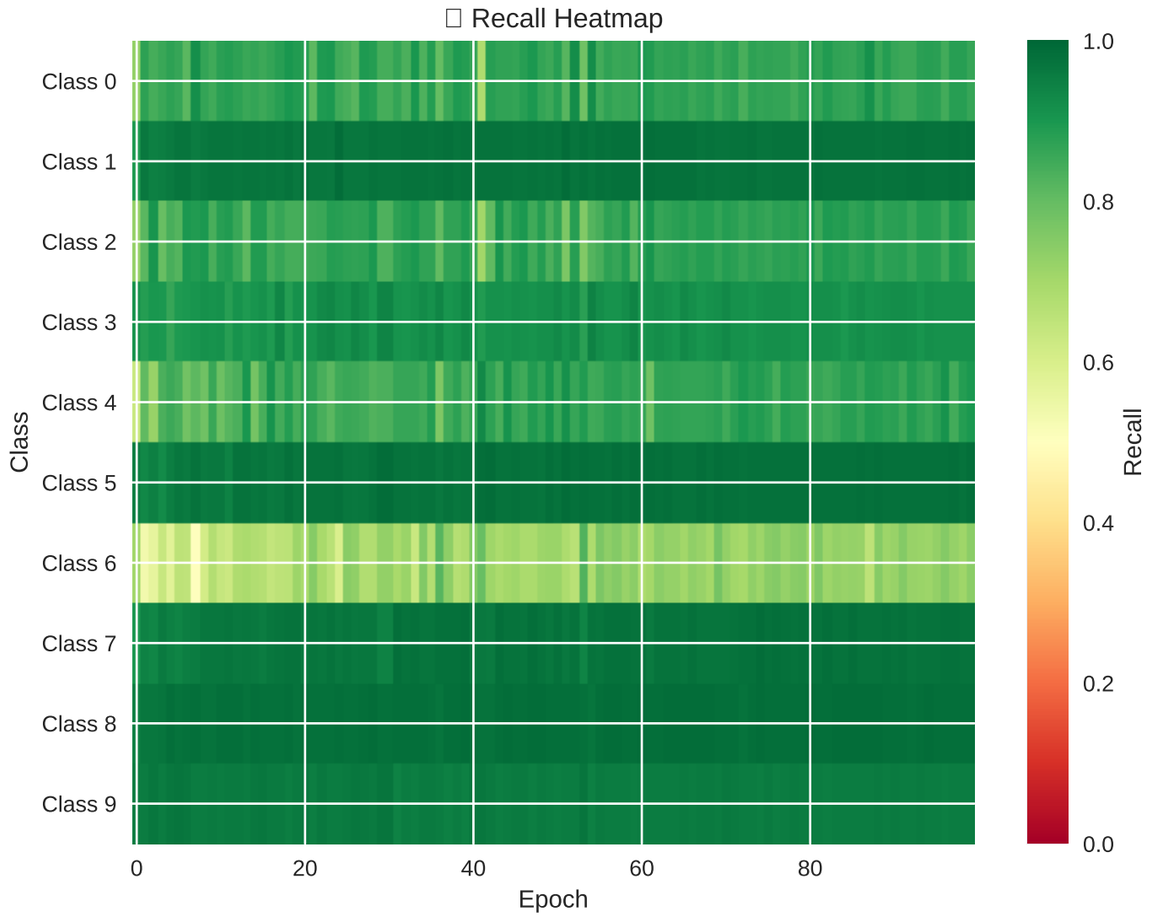


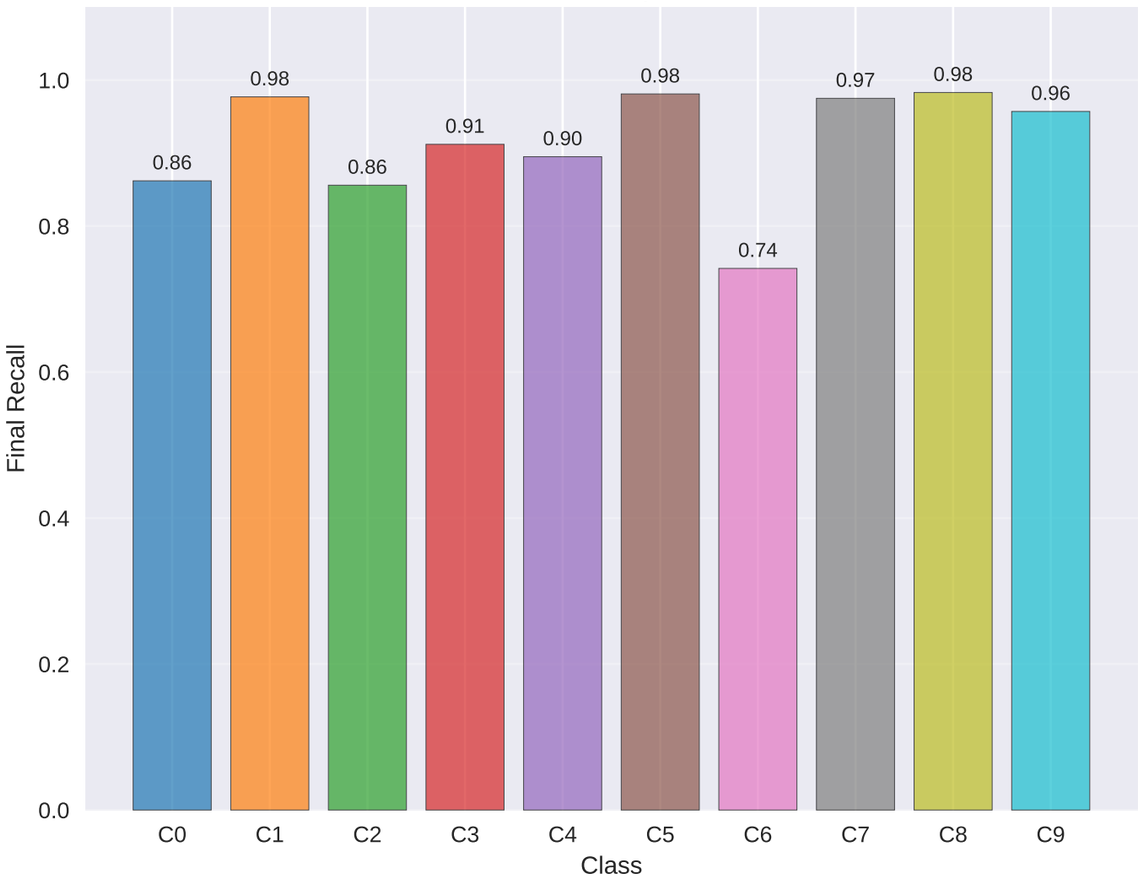


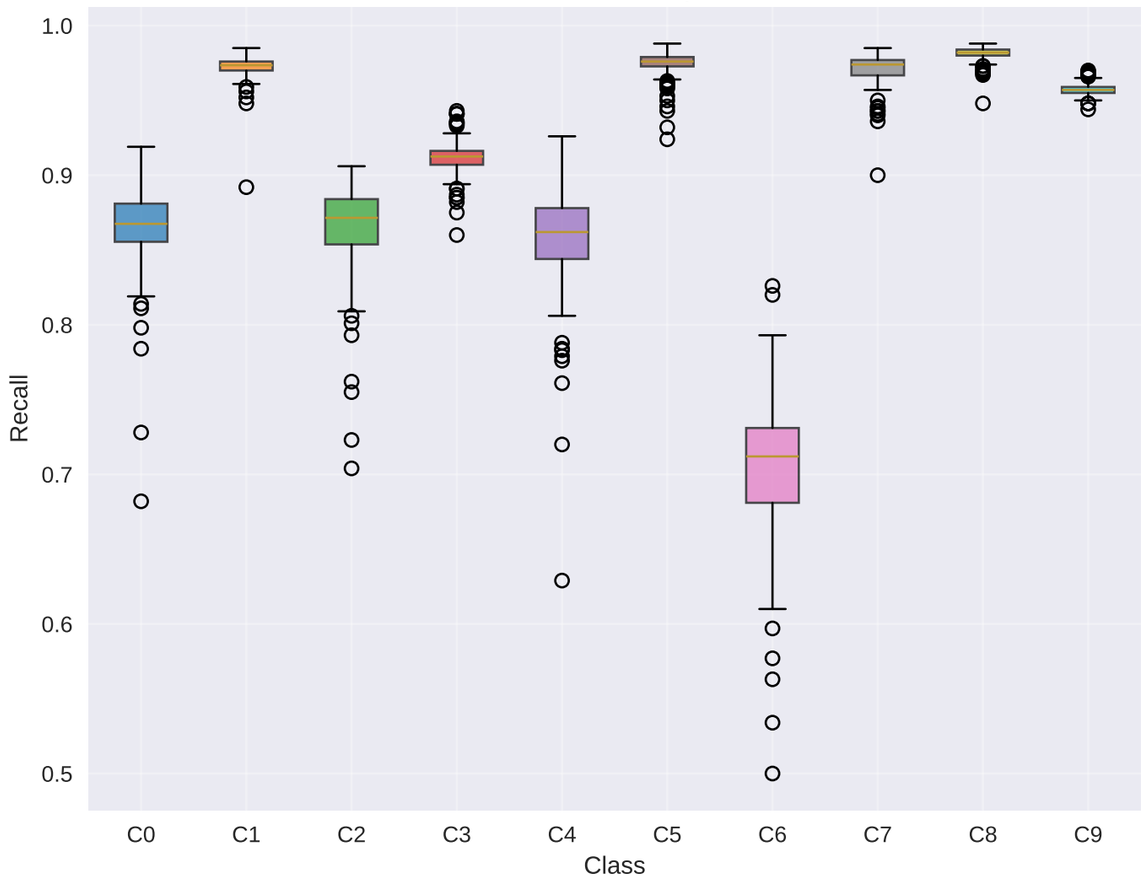


Per-Class Recall Analysis

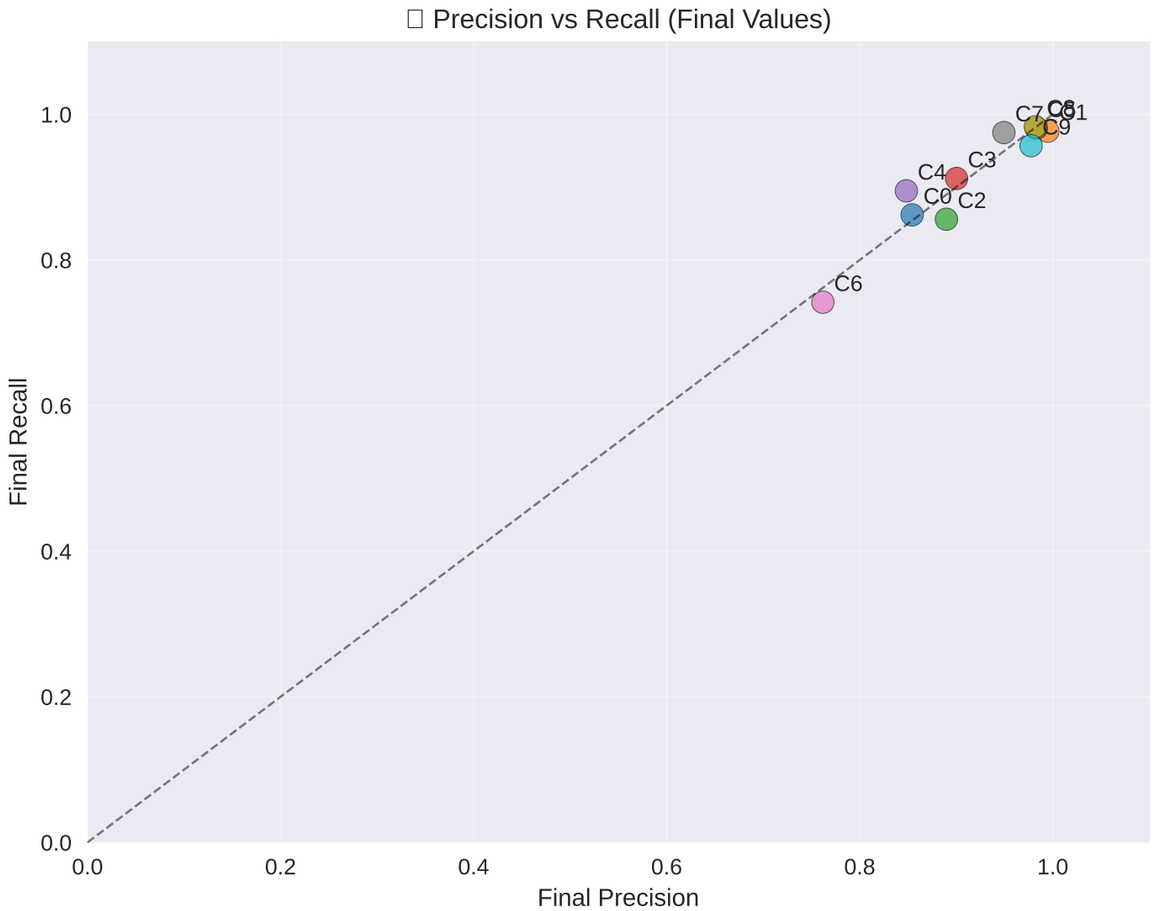






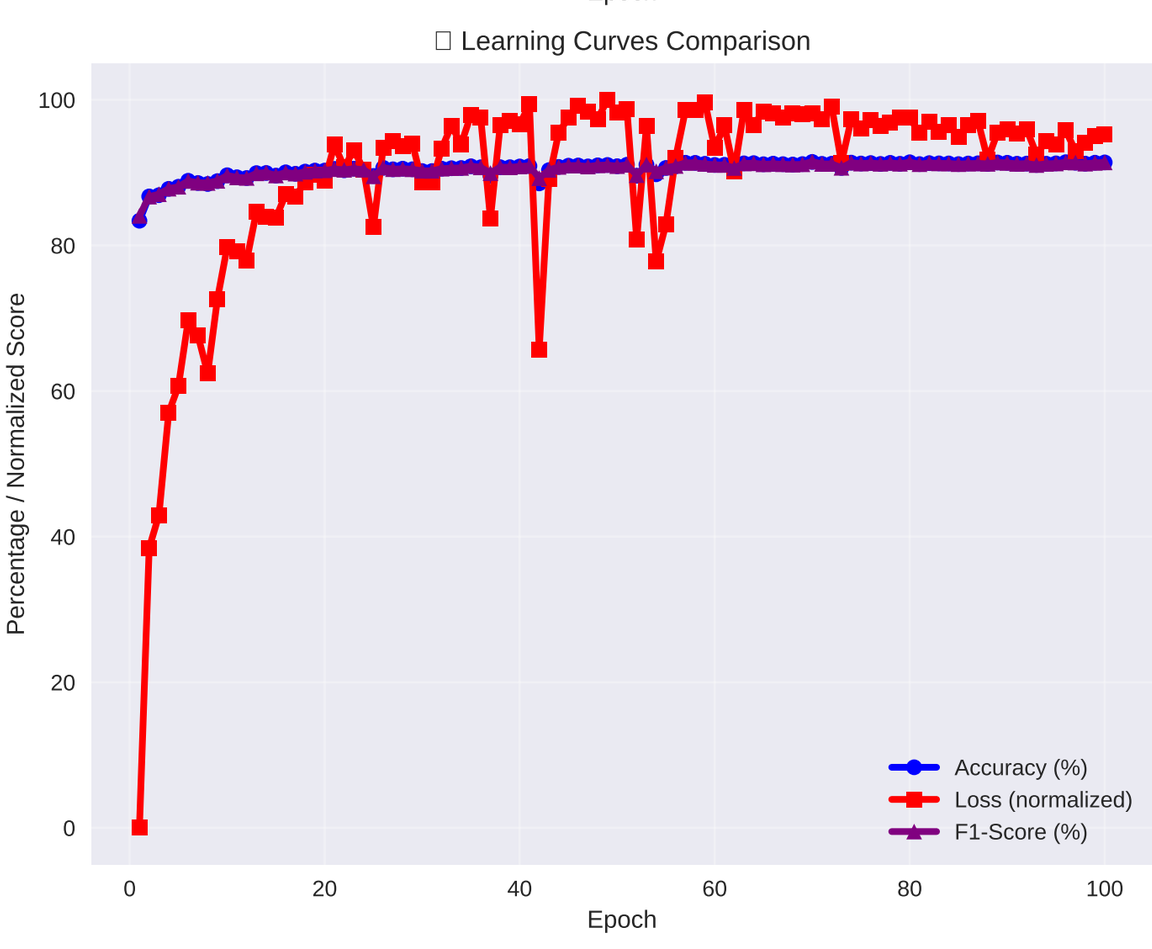


Comparative Analysis









Performance Summary Report

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FEDERATED LEARNING - PERFORMANCE SUMMARY
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FILE ANALIZZATO: 3000_results.csv
DATA ANALISI: 2025-08-31 00:41:12
EPOCHE TOTALI: 100

METRICHE FINALI:
• Accuracy: 91.40%
• Loss: 2.6162
• Average Precision: 0.914
• Average Recall: 0.914
• Average F1-Score: 0.914

EVOLUZIONE TRAINING:
• Miglioramento Accuracy: +8.01%
• Riduzione Loss: +1.9668
• Stabilità Accuracy (CV): 0.0134

PERFORMANCE PER CLASSE:
• Migliore Precision: Classe 1 (0.995)
• Peggiore Precision: Classe 6 (0.762)
• Migliore Recall: Classe 8 (0.983)
• Peggiore Recall: Classe 6 (0.742)

ANALISI DETTAGLIATA PER CLASSE:
• Classe 0: P=0.854, R=0.862, F1=0.858
• Classe 1: P=0.995, R=0.977, F1=0.986
• Classe 2: P=0.890, R=0.856, F1=0.873
• Classe 3: P=0.900, R=0.912, F1=0.906
• Classe 4: P=0.848, R=0.895, F1=0.871
• Classe 5: P=0.983, R=0.981, F1=0.982
• Classe 6: P=0.762, R=0.742, F1=0.752
• Classe 7: P=0.949, R=0.975, F1=0.962
• Classe 8: P=0.982, R=0.983, F1=0.983
• Classe 9: P=0.978, R=0.957, F1=0.967

VALUTAZIONE QUALITATIVA:
• Performance Globale: BUONA
• Stabilità Training: STABILE
• Balance Classes: SBILANCIATO

RACCOMANDAZIONI:
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